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Financial Literacy: Identification of the Challenges, Needs, and Difficulties among Adults Living in Rural Areas

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Abstract: Financial literacy plays a crucial role in individuals' decision-making processes. The paper aims to thoroughly identify the financial literacy needs and challenges of adults with low financial literacy living in rural areas. The paper presents the results of a focus group study conducted among a target group (35 people) and interviews with financial experts (14 people) in the first quarter of 2023. The study allows us to identify common areas of financial knowledge crucial for adults with low financial literacy living in rural areas. The study was conducted in seven European countries as part of the Erasmus+ project "Learning by Experiencing Escape Rooms: Financial Literacy for Adults (FLER)". Key findings from the study revealed that participants exhibited low awareness of financial fraud and expressed a need to protect personal information when using digital platforms. There was a clear demand for improved knowledge in certain areas, such as budgeting, emergency funds, balancing risk and reward, and planning for retirement or long-term savings. Participants consistently viewed gamification as a valuable tool for enhancing their understanding of financial topics. The contribution of our study is that we narrow the research gap on the needs and challenges related to financial literacy among rural residents. The results will help pinpoint specific areas of financial knowledge and competencies that are particularly important for adults with low financial literacy. Moreover, these insights are crucial for developing educational content integrated into a virtual escape room to improve financial literacy through engaging, scenario-based learning.

Keywords: financial literacy; rural areas; adults; focus group

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1. Introduction

Financial literacy plays a key role in individuals' decision-making processes. The understanding of the concept of financial literacy needs to be traced back to the definition of fundamental or general literacy, as the idea of financial literacy is rooted in general literacy [1,2]. Historically, fundamental or general literacy was associated with alphabetic literacy, focusing on writing, reading, and arithmetic skills [3,4], closely connected to educational contexts and fundamental for knowledge acquisition [5]. Nowadays, the definition of literacy has evolved to include social skills, technological proficiency, and the ability to understand and interact with various forms of texts, including non-print texts [6,7].

The idea of financial literacy dates to the early 1900s with the advent of consumer education research and initiatives which began in the United States [8]. The conceptual definition of financial literacy is complicated since scholars and financial experts have long

disagreed on how to define this concept. Moreover, numerous scholars who have published studies about financial literacy have not plainly defined this concept [9].

The most common definitions of financial literacy are those which are provided by the OECD, i.e., "a combination of awareness, knowledge, skill, attitude and behaviour necessary to make sound financial decisions and ultimately achieve individual financial wellbeing" [10] or "knowledge and understanding of financial concepts and risks, and the skills, motivation and confidence to apply such knowledge and understanding in order to make effective decisions across a range of financial contexts, to improve the financial wellbeing of individuals and society, and to enable participation in economic life" [11]. Moreover, commonly used definitions are those provided by Lusardi and Mitchell [12] and Huston [13]. Lusardi and Mitchell [12] define financial literacy as "the knowledge of basic financial concepts and ability to do simple calculations". Huston [13] conceptualizes financial literacy as the knowledge of personal finance as well as the application of that knowledge. Summing up, financial literacy can be broadly understood as the combination of awareness, knowledge, skills, attitudes, and behaviors required to make sound financial decisions and achieve financial wellbeing. Definitions by the OECD, Lusardi and Mitchell, and Huston converge on these core aspects, namely understanding financial concepts, applying that knowledge in real-life contexts, and developing confidence in financial decision-making.

Financial literacy is also connected to other concepts, such as financial capability, education, and awareness [14]. Financial literacy involves a combination of awareness, skills, knowledge, attitude, and behaviors necessary for sound financial decision-making [15,16]. Based upon a review of research studies of the many conceptual definitions of financial literacy, Remund [17] classifies them into four categories, i.e., knowledge of financial concepts, ability to manage personal finances, skill in making financial decisions, and confidence in future financial planning. Knowledge of financial concepts refers to general financial knowledge [18]. The definition of ability in managing personal finances involves keeping track of cash resources and payment obligations, knowledge of how to open an account for saving and how to apply for a loan, a basic understanding of health and life insurance, the ability to compare competing offers, and planning for future financial need [19]. Skill in making financial decisions is defined as the ability to make informed judgments and to make effective decisions regarding the use and management of money [20]. Finally, confidence in future financial planning refers to understanding about investing and financial planning [21]. The primary objective of the study is to gain a comprehensive understanding of the specific needs and challenges faced by adults from rural areas in relation to financial literacy. The target group embraces adults living in rural areas, without higher economic or financial education, and self-identified as not having a high level of financial literacy.

According to Amari & Jarboui [22] inadequate levels of financial literacy, particularly among young adults, is a global problem. Van Rooij et al. [23] find that financial literacy affects financial decision-making. James et al. and Klapper et al. [24,25] established that higher health and financial literacy levels correlate with improved decision-making abilities among older individuals in community settings. This suggests that enhancing literacy levels could lead to better decision-making, thereby improving health and life quality in later years. De Bassa Scheresberg [26] found that young adult individuals with greater financial literacy or more confidence in their knowledge of mathematics and personal finance tend to experience more favorable financial outcomes. Adults with high self-assessed confidence in managing personal finance but low financial literacy have a higher propensity to engage in undesirable financial behaviors [27].

Financial literacy has been found to be negatively associated with high-cost borrowing behaviors among individuals [28]. Low levels of financial literacy often lead to higher levels of debt, lower credit scores, less wealth accumulation, and poor retirement planning among individuals, especially among lower-income individuals in urban communities [29]. Financial literacy is associated with household asset allocation, with individuals with higher literacy also having investments in stocks or securities [30]. Low financial literacy

has implications for individuals, communities, countries, and society as a whole, leading to failure to plan for retirement, lack of participation in the stock market, and poor borrowing behaviour [31]. Demographic and socioeconomic attributes, such as gender, age, education, and income, influence financial literacy among adults [32,33]. Adult males and females have different financial socialization experiences associated with varying financial literacy and confidence levels. Education in finance and economics is linked to higher financial literacy and confidence in managing personal finances [27].

Pham and Le's [34] findings suggest that enhanced financial education and literacy significantly improve financial behaviors in young generations, emphasizing the need for targeted educational interventions. Klapper and Lusardi [25] assessed financial literacy among adults by evaluating their understanding of four key concepts crucial in financial decision-making, including knowledge of interest rates, compounding interest, inflation, and risk diversification. Their findings revealed that only about one-third of adults globally demonstrate financial literacy. Schuhen et al. [35], assessing the financial literacy of adults in Germany, revealed a general lack of financial knowledge across the population. The study calls for comprehensive financial education programs to address these deficiencies and improve financial decision-making. Méndez et al. [36], focusing on designing and evaluating a financial literacy scale for young adults, found that the validated scale offers a reliable tool for measuring financial literacy and can be used to develop effective financial education strategies.

As was mentioned above, we are focused on adults living in rural areas. Several studies highlight that individuals residing in rural areas exhibit lower financial literacy than urban financial consumers [37–40]. Financial literacy is critical for making informed financial decisions, such as managing debt, saving for retirement, and understanding loans. In rural areas of developed countries, lower financial literacy can lead to poor financial decisions, such as taking on high-interest loans or not saving adequately for long-term goals. This can exacerbate financial instability and increase the likelihood of financial distress [41]. Age, income, and education level positively impact financial literacy among adults in rural areas [42,43]. The research also indicates a significant gender gap in financial literacy in rural areas, with a lower percentage of females being financially literate than males [35,44]. Governments and regulatory bodies are running financial literacy campaigns, and there are recommendations to include financial education in school and college curricula to enhance financial inclusion [45]. Xu et al. [46] found that better financial literacy enables farm households to diversify income sources and improve financial stability, highlighting its importance for rural development. Financial literacy in rural areas supports the development of small businesses by equipping entrepreneurs with the skills to manage finances effectively, secure investment, and optimize business operations. This is particularly important in rural areas, where small businesses often play a central role in the local economy and job creation [47]. Liu et al. [48] concluded that higher financial literacy reduces credit constraints of rural households and influences entrepreneurial activities positively. Thus, in rural areas, where access to financial education and resources is limited, the relevance of these financial literacy components is even more pronounced.

Even in developed countries (as our study focuses on developed European member states), rural populations often face lower financial literacy levels, which affect their financial decisions and overall economic wellbeing. Nicolini et al. [49], comparing financial literacy in Canada, Italy, the UK, and the USA, found that rural populations consistently score lower on financial literacy tests than urban populations in these developed countries. Moreover, Nicolini [50], assessing financial literacy across several European countries, revealed disparities in knowledge across different areas of financial literacy (e.g., money management, investments). This highlights how rural areas tend to have lower scores, particularly regarding investment products. A study comparing rural and urban populations in Visegrad Group countries found that only 43% of rural residents achieved the minimum target score in basic financial knowledge, indicating significant gaps in understanding

financial concepts [51]. In Italy, a study by Banca d'Italia revealed that there is a substantial financial literacy gap between rural and urban areas, with lower levels of financial knowledge in rural regions. This disparity was most evident among older adults, women, and those with less formal education [52]. In rural areas of the U.S., for instance, older adults often exhibit responsible payment practices but struggle with using more complex financial products, which limits their financial wellbeing [53]. Studies emphasize the importance of targeted financial education programs in rural areas of developed countries. Financial literacy programs, particularly those tailored to address the needs of older adults and women, have been shown to significantly improve financial outcomes, encouraging better financial decision-making and resilience [54].

This proves that the problem of insufficient level of financial literacy concerns both developing and developed countries. Our study, as a part of the FLER project, responds to the challenges facing the EU countries, as one of the goals of the European Commission is to work to ensure that people throughout Europe have the knowledge and skills they need to make good financial decisions. Financial literacy is vital for enhancing personal financial management skills, which contribute to overall economic stability [55]. Additionally, it empowers individuals to participate in formal financial markets, enhancing financial inclusion [56]. Moreover, financial literacy plays a key role in reducing socioeconomic inequalities [57].

Financial literacy for adults living in rural areas is crucial for several reasons. Firstly, access to financial courses in these regions is limited due to the greater distance from higher education institutions. Rural populations often have more limited access to education, which correlates with lower financial literacy. Studies show that higher education levels are associated with better financial knowledge and skills [42,57]. Moreover, financial institutions and services are less prevalent in rural areas, reducing exposure to and experience with formal financial products. This lack of access contributes to financial exclusion and limits opportunities to develop practical financial skills [58,59].

The literature review clearly indicates that there is a lot of research on financial literacy among adults. Still, these studies focus on assessing the level of knowledge, skills, and financial competencies and indicating factors, including social, economic, and demographic factors, that determine and differentiate the level of financial literacy among adults. In addition, the research focuses on the impact of the level of financial literacy on individual households and entire regions or countries. The review highlighted the lack of research aimed at identifying areas of financial knowledge that are key from the perspective of an adult who does not have (or has a low level of) financial literacy. Additionally, due to the more significant problem regarding the level of financial literacy in rural areas, our study focused on this population. Additionally, when identifying the challenges, needs, and difficulties among adults living in rural areas, we examine not only the target group but also survey financial experts to have a more holistic approach. This study aims to fill that gap by addressing the unique financial challenges and needs of this underserved demographic, contributing to a broader understanding of financial literacy in rural contexts. The results will help pinpoint specific areas of financial knowledge and competencies that are particularly important for adults with low financial literacy, highlighting the greatest needs in this regard. Our study might contribute to developing targeted strategies to improve financial literacy in rural areas, which could inform policies, educational programs, and interventions to promote financial inclusion for this population. It aims to substantially contribute to this research gap, addressing an underexplored problem. Our research offers a broader and more valuable perspective by covering seven countries. This multi-country approach allows for cross-country comparisons, offering insights into the common and differing financial literacy challenges faced by rural adults across various cultural and socioeconomic environments. This comparative analysis might enhance the study's generalizability and practical relevance. Furthermore, according to 'D'Alessio et al. [60], financial literacy varies among different countries.

The outline of this paper is as follows: the next section presents the overview of the Erasmus+ project FLER (Learning by Experiencing Escape Room: Financial Literacy for Adults) and provides the methodology of the research. The third section details the results of our study. The final section contains the conclusions and recommendations for further investigating the problem.

2. Materials and Methods

The research is conducted within the Erasmus+ Learning by Experiencing Escape Rooms: Financial Literacy for Adults (FLER) project. The general objective of the FLER project is to equip low financially literate adults belonging to rural areas with a specific digital learning tool (a virtual escape room) aimed at hosting a scenario-based learning course on financial literacy, which will represent an effective learning approach, providing users (i.e., learners) with a realistic context and emotional engagement, thus allowing for a more efficient understanding and retention of financial topics. A virtual escape room is an online interactive game in which participants solve puzzles, uncover clues, and complete different tasks within a set time to escape from a virtual environment (room). The application of virtual escape rooms enhances learning by promoting problem-solving, critical thinking skills, and teamwork. By simulating real-world challenges, they help learners apply theoretical concepts in a practical and enjoyable context. The FLER project's consortium consists of the following seven institutions, including the coordinator, the Warsaw University of Life Sciences (WULS-SGGW) from Poland, Die mitra Education & Consulting S.A. (DIMITRA) from Greece, Finance & Banking, Associazione per lo Sviluppo Organizzativo e delle Risorse Umane (EFFEBI) from Italy, Eurodimensions from Malta, Folkuniveristetet Uppsala from Sweden, Fundació Privada Institut d'Estudis Financers (IEF) from Spain, and the Virtual Campus from Portugal. The project aims to equip adults from rural areas with financial knowledge and skills. There is a high need to build effective and attractive learning tools that allow people to acquaint themselves with fundamental economic categories, understand the rules of making financial decisions, understand the types of financial fraud and the methods to avoid them, to prepare and control the household budget, to assess the profitability of an investment, and to develop the skills of communication tools with a financial institution.

The focus group meetings and interviews with experts were one of the first tasks in the FLER project, and these allowed us to identify the needs, challenges, and problems of adults with low financial literacy. We employed a qualitative research design, which allowed us to capture in-depth insights into the target group's needs. The combination of focus groups and expert interviews was selected to gather a broad range of perspectives, with focus groups allowing for interactive discussions and expert interviews providing specialized knowledge.

The study was divided into two parts. First, we conducted focus group research among adults residing in rural areas. Participants for focus groups were selected using purposive sampling, ensuring the inclusion of adults with low financial literacy from rural areas. Subsequently, we discussed the results obtained in the focus group studies with experts. Experts were chosen based on their professional experience in financial literacy and adult education, ensuring relevance to the study's objectives. Participants were recruited through local partner organizations involved in the FLER project. Several benefits emerge when considering the advantages of employing the focus group method and conducting interviews with experts in our study. Firstly, focus group research allows for in-depth exploration of participants' perspectives and experiences, fostering a more comprehensive understanding of the subject matter. Additionally, the interactive nature of focus groups promotes dynamic discussions that can reveal nuanced insights. On the other hand, engaging with finance experts through interviews provides specialized knowledge and insights, contributing to the depth and accuracy of the study. Overall, combining these methods enhances the richness and validity of the research findings.

A focus group is a gathering of deliberately selected people who participate in a planned discussion intended to elicit stakeholders' perceptions about a particular topic or area of interest in a receptive environment. Focus groups allow group members to interact and influence each other during the discussion and consideration of ideas and perspectives. Basically, focus groups are a way to reach stakeholders for feedback and comments. According to the project, each focus group consisted of at least five carefully recruited participants (at least three low literate adults coming from rural areas and two financial literacy/adult education and training experts/associated partners) from each country (IT–PL–SP–MT–SE–PT–EL). Altogether there were 35 focus group participants. The participants were aged 18–65; among them were farmers, permanently employed professionals, individual entrepreneurs, unemployed individuals, retired individuals, and refugees residing in rural areas. At the beginning of the focus meetings, low financially literate participants completed a questionnaire. The attained results indicated that the financial knowledge of the target group varied depending on the country and age of the participants. Nonetheless, all focus group participants (in all countries, i.e., PL, ES, MT, PT, IT, EL, and SE) indicated the need to broaden their financial knowledge and address the main challenges in the proposed thematic areas.

In the second part of the research, experts were invited for individual interviews from the group sessions in each partner organization. Discussions were structured around consecutive questions asked by the interviewer (which were unified for all countries) and aimed to detect experts' opinions on the existing needs and challenges low financially literate adults living in rural areas may encounter. Based on personal and professional experience, participants were asked to describe the main obstacles in identifying adults' needs on proposed topics from their point of view. Furthermore, experts shared their opinions about the advantages and disadvantages of developing an engaging learning digital approach and its usefulness for end-users to raise the effectiveness of understanding and retention of financial topics.

Focus groups and interviews followed a standardized protocol across all seven countries, with sessions lasting approximately 90 min and being conducted in accessible locations. Discussions were guided by a semi-structured questionnaire unified across all participating countries. Data were recorded using a combination of audio recordings and detailed note-taking. Informed consent was obtained from all participants. Analysis was carried out both individually and collectively to identify commonalities and differences across countries.

The research addresses the following two questions:

- What are rural adults' specific financial literacy needs and challenges in their own opinion?
- What are the specific financial literacy needs and challenges faced by rural adults in the opinion of financial experts?

The paper aims to identify the financial literacy needs and challenges of low financially literate adults from rural areas. To achieve this objective, a qualitative study was conducted, including focus group discussions and expert interviews in seven European countries. The method was explicitly chosen to address the study's aim, with focus groups identifying real-world challenges and expert interviews providing deeper insights. The findings obtained through the research have shed light on the specific areas where adults in rural areas face difficulties in understanding and managing their finances. These insights are instrumental in developing tailored educational content that addresses the identified needs and challenges.

3. Research Results and Discussion

Considering the study's methodology, the survey results are presented separately for the low financial literacy of adults from rural areas and financial experts. For both interviewed focus groups, Tables 1 and 2 have a common structure, comprising the list of evaluation criteria for particular thematic areas, their description, and the main findings for the relevant area.

Table 1. Needs and difficulties related to financial literacy in opinions of adults from rural areas.

Criteria/Thematic Area	Description	Main Findings
Focus group description	The panel consisted of people of different age groups, backgrounds, and professions from rural areas who declared no prior education or training in finance and expressed their interest in enhancing their financial literacy.	The participants were 18–65; among them were farmers, permanently employed professionals, individual entrepreneurs, unemployed individuals, retired individuals, and refugees residing in rural areas.
A. Self-assessment of financial literacy	In the first section of the questionnaire, adults from rural areas were asked to provide a general assessment of their financial knowledge and rate their degree of interest related to a virtual escape room tool on financial literacy.	The majority of the focus group rated their financial literacy as low and insufficient for making financial decisions. Participants indicated their strong interest in taking various forms of financial literacy training, including virtual escape rooms on financial literacy. Moreover, it was pointed out that older participants were more concerned about their abilities to work with new interactive educational instruments, such as e-learning courses and escape rooms, due to their limited computer skills. Employed and unemployed and businessmen mentioned the necessity of enhancing financial literacy for pursuing their professional goals.
B. Testing financial knowledge	In the second section of the survey, participants were asked to take a short test that consisted of three questions on basic financial literacy knowledge related to the time value of money, inflation, and financial investment concepts.	The quick financial knowledge test results confirmed the need to improve adults' financial knowledge and competencies. Many participants do not understand the concept of the time value of money and its applications; there is a lack of general knowledge on the investment decision-making process, particularly diversification to minimize the investment risk.
C. Dimensions of financial literacy	In the third section of the questionnaire, the interviewer tested the competence of focus group participants on five different topics of financial literacy. The main results are summarized in the presented rows below (I–VI).	
I. Financial fraud awareness	In this section, respondents assessed their awareness and ability to resist financial fraud, including concerns about personal data protection in communication with financial institutions and intermediaries and the safety of electronic and digital channels for payment.	Most respondents assessed their awareness of financial fraud as low and pointed out that they had insufficient competencies to resist the associated risks. The main threats included the inability to find reliable information on the types of financial fraud, the failure to protect against personal data stealing, suspicious mailings, phishing scams, cyberattacks, and telephone and online fraud. Participants highlighted the importance of Internet information campaigns on financial fraud and their eagerness to enhance their knowledge on related topics.
II. Budgeting and emergency funds	This section concerned the issues of personal finances, including budgeting and such concepts as income, costs, cash management, and short-term savings.	Most participants' budgeting and personal financial management knowledge is moderate or low. Many respondents do not plan their spending or have an emergency fund. However, the majority of participants indicated the availability of savings for a period from two to six months. Older participants declared a higher cautiousness on the issues of savings and emergency funds than their younger counterparts. Respondents highlighted the significant role of crediting institutions in providing emergency financing.

Table 1. *Cont.*

Criteria/Thematic Area	Description	Main Findings
III. Balancing risk and reward	In this section, respondents evaluated their understanding of financial risk, its sources and factors, and the instruments used to mitigate the risks. The survey paid particular attention to insurance and its cost.	The participants demonstrated limited knowledge of the financial risk concept and the relationship between risk and reward. However, some respondents could identify the major types of risks and their sources and classify the financial instruments for risk management. Notably, most participants were familiar with or used different types of insurance, including life and asset insurance. The awareness of insurance mechanisms increased in the regions and groups with a higher income.
IV. Planning for retirement or other long-term savings	This section focused on long-term financial management, including building a retirement plan and constructing a long-term financial strategy.	The younger respondents' knowledge of long-term financial management and retirement plans was very limited. Older respondents demonstrated a good understanding of pension and retirement plans and underlined the necessity for long-term savings. In their financial strategies, the respondents seem to rely on more 'sustainable' instruments, such as long-term investment in real estate or saving money in bank deposit accounts.
V. Investment products	The section evaluated the respondents' knowledge of investment and the investing process, the concept of risk diversification, and the portfolio structure.	Respondents demonstrated limited knowledge of investments and the investing process. Some participants were able to identify the main financial instruments, including stocks, bonds, and loans, as well as to underline the role of the interest rate in the investing process. In their financial strategies, most respondents preferred investing in less risky instruments, such as bank deposit accounts, compared to the high risk of investing in stocks and derivatives.
VI. Other dimensions	Respondents were asked about other areas of interest in finances.	Respondents expressed their interest in improving their knowledge of taxes and taxation, as well as of macroeconomic and corporate financial indicators and ratios.

Source: own research.

The interview results with the adults from rural areas are presented in Table 1. It includes the main findings of self-assessment of financial literacy, tests of financial knowledge, and the extended survey on various dimensions of financial literacy, such as financial fraud awareness, budgeting and emergency funds, balancing risk and reward, planning for retirement and other long-term savings, investment products, and other dimensions not covered by the scope of the interview. In selecting the thematic areas, we followed the financial competence framework for adults in the European Union [61].

The results of the interviews in the focus group highlighted the general need for enhancing the financial literacy among the inhabitants of rural areas, which was in line with a broad academic view [62,63]. Most participants rated their financial literacy level as insufficient for making sound financial decisions and demonstrated a strong interest in developing financial competencies using modern interactive tools, including virtual escape rooms. For example, according to the focus group in Poland, financial literacy is essential for both personal and professional purposes. The participants rated their knowledge of relevant areas as low but also stated their willingness to expand financial competencies. Furthermore, the majority of the Polish participants indicated that they had never participated in training related to finances (on-site), and about half of them had never attended any interactive courses. Similarly to Poland, in Spain, most of the respondents did not have a sufficient level of financial knowledge and declared they had never participated in an escape room as a form of education. Moreover, both groups pointed out that gamification and virtual escape rooms represent an

attractive and effective way to increase the level of financial knowledge. Some participants claimed that schools should introduce financial literacy courses to encourage responsible behaviour patterns among children and teenagers. Notably, the interest in specific financial literacy thematic areas shifted depending on the respondents' age, background, and previous experience, confirming the importance of individuals' heterogeneity in approaching financial issues [64,65]. The results of a provided self-assessment were supplemented by short financial literacy tests among the focus group representatives, confirming the general lack of knowledge on financial concepts, including the time value of money, investment and budgeting processes, and risk management. These findings emphasized the need to introduce financial education and to enhance financial literacy in rural areas [66,67].

Table 2. Needs and difficulties related to financial literacy in opinions of financial experts.

Criteria/Thematic Area	Description	Main Findings
1. Diagnosis of financial literacy deficiencies among adults	In the section, experts investigated main financial matters where low financially literate adults living in rural areas lack proper knowledge.	Among the main financial matters to be both actual and ambiguous for low financially literate adults, the experts diagnosed emergency fund and budgeting, investments, credits, access to grants and subsidies, work-related insurance and pensions, and the risk of financial fraud.
2. Recommendations on enhancing financial literacy	This section contains recommendations to enable low financially literate adults living in rural areas to make sound financial decisions.	The main recommendations included taking organized courses, workshops, or training to acquire the necessary knowledge and skills. The proposed topics of learning activities were financial management, budgeting, pension and retirement plans, grant acquisition and management, investing in various financial instruments, and risk analysis. The importance of self-education was also highlighted.
3. Obstacles to developing digital skills	In the section, experts discussed challenges/problems that low financially literate adults living in rural areas may encounter when trying to enhance their digital skills.	The experts pointed out different obstacles to developing digital skills, such as recipients' low motivation to learn, insufficient previous backgrounds, and language barriers, as well as the lack of necessary equipment, limited access to infrastructure and technology, and scarcity of financing to bear education initiatives. Experts' opinions were subject to the geographical distribution that the national context and the overall level of regional digital and infrastructure development can explain.
4. Instruments for utilizing digital skills to improve financial literacy	This section focused on opportunities deriving from the enhancement of the digital skills for low financial literacy adults coming from rural areas	Experts stressed the crucial role of institutional support and digital channels in enhancing financial literacy, such as mobile applications, broadcasting and learning digital platforms, social media, virtual reality, and desktop software. They pointed out the emerging function of gamification, including virtual escape rooms and puzzles, in improving financial literacy.

Source: own research.

The participants assessed different dimensions of financial literacy and demonstrated moderate or little experience in particular fields. In the financial fraud awareness thematic area, the respondents correctly recognized the threats arising from financial frauds and scams, signaling the need to protect personal information on digital platforms, despite some struggling with new technologies. Participants identified such threats as phishing, malware and spoofing attacks, card cloning, online investment frauds, and fake broker agents. Furthermore, participants reported a strong demand for more experience in using online and mobile banking channels, underlining the important role of social media and information campaigns in delivering the appropriate knowledge and training. Generally, this provided evidence of the increasing significance of digital technologies from the

perspective of users of financial products, justifying the necessity for conducting more extensive training in related fields among European residents of rural areas [68–70].

In the budgeting and emergency fund area, the participants' limited engagement in budgeting activities underscores a fundamental gap in their financial literacy [71]. Budgeting is a cornerstone of financial management, yet the majority of focus group members did not incorporate it into their household finances, indicating a lack of understanding of basic financial planning principles. The majority of respondents estimated their budgeting skills as low or moderate, largely focusing on the operating expenses of a household. Awareness of the benefits of savings increased with the age of participants, indicating that younger respondents were less interested in accumulating funds for future expenses. This fact reflected a general trend of changing budgeting strategies based on individual or household characteristics [72]. Generally, in the national focus groups, many participants claimed that low income was the main reason preventing them from creating their personal emergency fund, confirming the results of other studies on the relationship between financial literacy and the financial wellbeing, [73]. For example, in Portugal, participants were not in the habit of planning household budgets or tracking expenses, as most stated they could survive for six months without their main income, while in Spain, this period varied greatly from several weeks to several years. Therefore, their inability to establish emergency funds due to low income highlights low financial resilience and exposure to financial distress [74].

In balancing risk and reward, respondents were aware of the significant risks arising from investing in various financial instruments, such as stocks and currencies, while financial risk was often associated with global capital funds. In mitigating the risks, many respondents identified different types of insurance as the main mechanism, gaining significance with personal income growth. In Italy, respondents identified external risks impacting financial wellbeing, such as earthquakes for the merchant, bad weather for the farmer, and car accidents for the commercial agent. Simultaneously, most of the respondents in different countries demonstrated an emerging interest in risk diversification and acquiring competences in risk management. However, considering the limited knowledge of the risk–reward mechanisms and the rising complexity of financial markets, the participants' risk measurement, evaluation, and comparison skills were limited [75].

In planning for retirement or other long-term savings areas, there was a clear relationship between the age of respondents and their eagerness to manage finances 'sustainably' in the long term. This confirmed the fundamental view on the existing nexus between demographics and financial behavior patterns [76]. For Poland, Portugal, and Sweden, with a younger population of respondents, there was a lack of understanding and interest in pension plans, largely focusing on long-term investments in the form of real estate or traditional business projects rather than retirement savings. In Greece, respondents acknowledged the importance of planning for retirement or other long-term savings but did not understand the procedures for creating one. In farming communities of Italy, middle-aged participants had a good perception of retirement plans, with one relying on a broker for long-term savings. These differences could also result from a country-level specificity of pension systems and the profiles of respondents [77].

The investment products area proved the high investment risk aversion among the majority of respondents mainly due to insufficient financial knowledge and a cautious approach to financial instruments [78]. For example, in Poland, participants claimed their understanding of financial products, like shares and bonds, was superficial, and they had difficulty learning about currency price fluctuations and identifying profitable investments. Similar results in other countries indicated that participants, on average, had basic knowledge of stocks, interest rates, savings certificates, foreign exchange, and bonds, and preferred saving money over investing, viewing stocks as riskier but more profitable. However, younger people demonstrated interest in acquiring appropriate competencies in stocks, interest rates, cryptocurrencies, NFTs, foreign exchange, and bonds. Additionally, participants expressed distrust in traditional investment channels and financial institutions, and their preference for alternative investment avenues, such as small and volatile

business activities, suggested a lack of understanding of investment fundamentals and risk assessment on the one hand and the lack of sufficient funds to diversify on the other. However, the reluctance to invest in well-known financial products, despite awareness of them, reflected a deeper misunderstanding of investing strategies and potential returns. Moreover, participants demonstrated a limited understanding of financial indicators, such as stock market indexes and earnings per share, further deepening the gap in their risk aversion. These findings proved the urgent need for increasing individuals' investment proficiency to combat extreme investment behavioral patterns in a highly turbulent financial environment [79,80].

For other dimensions of financial literacy, respondents indicated the necessity of learning more about the interactions between different economic factors and fiscal policies influencing personal finances and investment decisions.

To sum up, the focus group's responses indicated a pervasive deficiency in financial education and awareness, emphasizing the critical need for comprehensive financial literacy initiatives tailored to their specific needs and circumstances. The quick financial knowledge test results confirmed the need to improve adults' financial competencies. Participants lacked knowledge of core financial concepts, including the time value of money, investments, and risk. They also wrongly interpreted basic types of financial instruments, such as investment in stocks, bonds, or deposits, debit and credit cards. With regard to financial fraud, participants expressed that they lack awareness and identified the need to protect personal data when using digital platforms. Moreover, respondents indicated the need to increase awareness and education on fraud prevention. Phishing scams and malware attacks were commonly identified as significant. Considering budgeting and emergency funds, participants noted the need to understand household budget planning, counteract the inability to accumulate a financial cushion, improve budget decisions, and signal the general need to increase budgeting and emergency funds knowledge. With regard to income and risk, the focus group highlighted the preference for investing in traditional business activities and demonstrated moderate knowledge about insurance providers and products. Some participants pointed out concerns about the lack of trust in financial institutions and tools provided by banks, accelerated by market turbulence and increasing fiscal pressure. Other significant concerns revealed during interviews were retirement planning and long-term savings. The results of the research have shown that respondents were concerned about their future financial wellbeing and carefully investigated the opportunities of using various investment products. For example, participants expressed an interest in learning more about taxes related to their professional careers and, more generally, in improving their knowledge of the impact of fiscal and monetary policy on savings and investment income.

Table 2 presents the results of the interviews with the focus group of financial experts exploring the challenges in enhancing financial literacy among adults in rural areas. Participants diagnosed financial literacy deficiencies and formulated recommendations for bridging the gaps in it. Moreover, experts concentrated on the barriers to developing digital skills and possible instruments for utilizing them to improve financial literacy.

Based on first-hand experience, financial experts identified several critical components requiring particular attention in teaching financial literacy among inhabitants of rural areas. Supporting the evidence from the self-assessment and financial knowledge tests revealed in Table 1, professionals confirmed the adults' insufficient understanding of emergency funds and household budgets, investing, risk assessment and insurance, retirement planning, financial products including loans, grants, and subsidies, financial fraud, and growing financial bureaucracy. Experts suggested reducing the financial literacy gap by organizing training courses, the creation of educational service centers by local authorities and communities, raising awareness on understanding financial documentation, and eliminating digital exclusion by providing electronic devices and Internet access combined with educating in financial and digital proficiency. Moreover, they mentioned the crucial role of adults' commitment to self-education and development in enhancing financial competencies. The

introduction of these mechanisms should involve both regulatory and bottom-up incentives. First, regulatory incentives can provide a structured framework and enforce standards that drive compliance and promote widespread adoption [81,82]. Second, bottom-up incentives should engage and motivate individuals or organizations at the operational level, fostering a sense of leadership and encouraging active participation [83].

In the second part of the survey, experts outlined the potential advantages of improving digital skills for inhabitants of rural areas. These benefits included reducing digital exclusion, enabling access to financial knowledge and advisory services, fostering the use of digital financial instruments, and facilitating a better understanding of finance in business reality. In comments on the obstacles to developing digital skills on the path to mastering financial literacy, professionals emphasized the limited access to infrastructure, the high cost of necessary equipment, and the lack of trained educators in some rural areas. Moreover, they stressed that ambiguity of the overall level of economic development in different countries and regions might enhance the disproportionate access to digital channels and financial services. However, in the experts' opinions, regardless of spatial context, developing digital skills requires substantial institutional investments in digital communication channels, including creating desktop and mobile applications, the development of broadcasting and learning digital platforms, and the extensive use of social media and Internet educational campaigns. Admitting the importance of digital channels, respondents accentuated the emerging function of new approaches, such as gamification, including virtual escape rooms and puzzles, in improving financial literacy among adults in rural areas [84,85].

Our study revealed that adults lacking financial knowledge and skills are afraid of making financial decisions and tend to distrust financial institutions. This corresponds with the study of Van Rooij et al. [23], who found that financial literacy significantly influences financial decision-making. Focus group participants expressed concerns about their future financial security and showed a strong interest in learning more about retirement planning, taxes, and long-term savings. This is in line with the findings of Lusardi [31], who demonstrated that low financial literacy has significant implications for individuals, communities, countries, and society as a whole, resulting in inadequate retirement planning, limited stock market participation, and poor borrowing behavior. Our study supports Ramakrishnan [55], Batsaikhan & Demertzis [56], and Prete [57] who indicated that financial literacy components empower individuals to participate in formal financial markets, enhancing financial inclusion and overall contributing to economic stability and reduced socioeconomic inequality.

4. Conclusions

The purpose of the research was to better understand the needs and challenges faced by adults from rural areas in relation to financial literacy. The research employed a methodology involving focus groups meeting with adults with low financial literacy and interviews with adult education and training experts, financial literacy specialists, and associated partners. Adults from rural areas and experts were separately involved in two different sessions. The study is a part of the Erasmus+ Learning by Experiencing Escape Rooms: Financial Literacy for Adults (FLER) project. The general objective of the FLER project is to equip low financially literate adults belonging to rural areas with a specific digital learning tool (a virtual escape room) aimed at hosting a scenario-based learning course on financial literacy, which will represent an effective learning approach, providing users (i.e., learners) with a realistic context and emotional engagement. The findings presented in the paper are the driving point for the design of scenario-based learning courses on financial literacy for virtual escape room games.

The focus group research aimed to identify the needs and important issues related to financial literacy among adults living in rural areas. During the focus meeting, participants completed a short questionnaire at the beginning. The obtained results showed that the financial knowledge of the target group varies depending on the country of the study

or the age of the participants. However, all focus group participants indicated the need to broaden their financial knowledge. Importantly, across all countries (PL, ES, MT, PT, IT, EL, and SE), gamification was consistently evaluated as a valuable tool for enhancing understanding of financial topics. Below are the key conclusions derived from the focus group study conducted in the seven partner countries:

- Financial fraud awareness: Participants across all countries expressed low awareness of financial fraud, particularly in digital contexts, highlighting the need for education on protecting personal information.
- Budgeting and emergency funds: There was a clear need for improved understanding and skills related to budgeting, managing household expenses, and building emergency funds.
- Balancing risk and Rerard: Participants showed a preference for investing in business activities but lacked knowledge about insurance products, financial tools, and the implications of financial decisions. Additionally, they revealed a lack of confidence in financial tools provided by banks
- Retirement planning and savings: Concerns about future financial security were common, with participants expressing interest in learning more about retirement planning, taxes, and long-term savings.

As far as the interviews conducted with experts are concerned, they aimed to gather information about the experts' recommendations, challenges, and opportunities for enhancing financial skills in adults coming from rural areas. Also, experts' opinions on using a Virtual Escape Room as a digital learning tool were discussed. Additionally, the research helped establish connections with key stakeholders and experts who can support the project's development. The experts identified topics relevant for low financially literate adults living in rural areas, such as managing home budgets, investing, insurance, risk assessment, retirement planning, access to financial resources, and understanding documents from financial institutions. Among the biggest challenges for project success, experts mentioned the insufficient digital skills of the target group, low motivation, limited access to infrastructure and equipment, high costs, a lack of teachers, and practicality in everyday life.

The field research provided valuable insights into the financial literacy needs and challenges faced by low financially literate adults from rural areas. The findings were used to create educational content to be integrated into a virtual escape room to improve their financial literacy competencies through gamification. Additionally, the research helped to establish connections with key stakeholders and experts. The findings are instrumental in guiding the development of targeted educational content within the FLER project and offer a foundation for future efforts to improve financial literacy among rural populations.

Additionally, the finding might have several important implications for policy and enhancing financial competences. The research reveals the need for targeted financial literacy programs in rural areas, particularly those that address digital financial fraud and the unique challenges of rural economies. Gamification, through tools, like the virtual escape room, has been perceived as an effective method for engaging rural adults, especially young adults, in financial literacy education. This approach should be further explored and expanded.

We acknowledge some limitations of the study. The sample size may restrict the generalizability of the findings, and future research should consider larger samples. Additionally, the study was conducted in specific rural areas of seven European countries, which may not fully capture the diversity of rural experiences across Europe or globally. However, it is important to note that the study was conducted precisely with the target group defined in the FLER project, ensuring its relevance to the intended population.

The future research should focus on several key areas. First, it should explore the long-term impact of digital tools, such as the virtual escape room, on enhancing financial literacy among adults. Additionally, it is important to investigate the applicability of similar tools in different cultural and socioeconomic contexts to ensure their effectiveness across

diverse populations. Finally, examining the role of digital literacy in the success of financial education programs, particularly in rural areas, will provide valuable insights into how these programs can be tailored to meet the needs of various target groups.

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